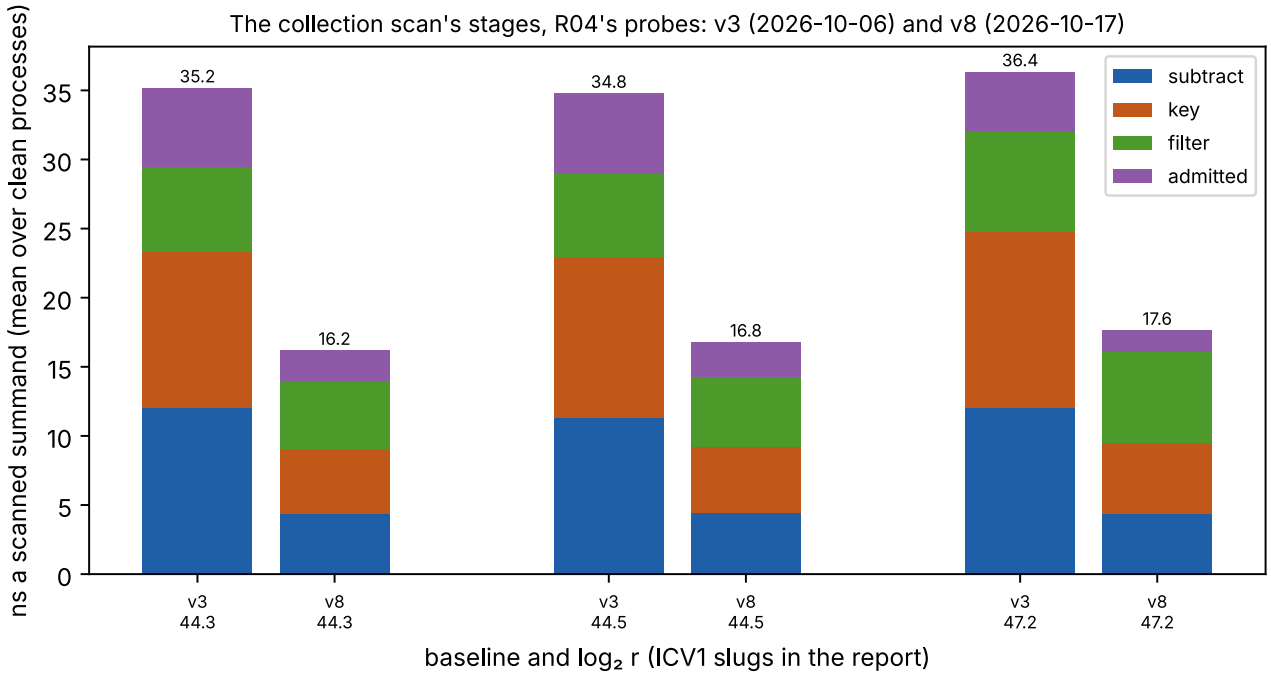
The scan's stages on v8: R04′s probes after R06, R08, R11 and R12

Stage diagnostic, no decision. R04′s probes (`--features scan-probes`), ported onto v8 (`23138d20`) as `786c06f7` (`probes-v8.patch`); the binary `ic-r04probes-on-v8-786c06f7`. Run 2026-10-17 03:37–03:43 UTC on the Emerald Rapids host R12 ran on: `M1`′s two rows at the three target sizes, four rounds, 24 isolated processes on CPU 2. Records: `research/ic_tool_program/explorations/scan-stages-v8-20261017/`.

- **Completed:** 24 of 24.
  - One process was contended (2^47.2, round 1) and is excluded.
  - The busy gate refused 13 starts, and each refused run was asked for again.
- **Logarithms:** every process recovered its row's logarithm, the one v8 recovers.
- **The port:** v8′s `admitted` stage includes R12′s run-prefetch pass, which v3 did not have.



explorations/scan-stages-v3-20261006/stages.tsv (2-4 processes a size); scan-stages-v8-20261017/stages.tsv (7-8); v8 counts R12's run prefetch in 'admitted'

Figure 1: ns a scanned summand per stage, v3 (`scan-stages-v3-20261006`) beside v8, means over the clean processes.

Stages, ns a scanned summand (v3 in brackets)

curve	$\log_2$ r	subtract	key	filter	admitted	scan
`icv1-f2m53-tm56619371-dac20a85`	44.3	4.35 (12.0)	4.70 (11.3)	4.95 (6.1)	2.19 (5.7)	16.2 (35.2)
`icv1-f2m59-tm943548413-98844ecc`	44.5	4.41 (11.3)	4.77 (11.6)	5.11 (6.0)	2.50 (5.8)	16.8 (34.8)
`icv1-f2m61-t158598901-ab42b6c5`	47.2	4.40 (12.1)	5.11 (12.7)	6.56 (7.4)	1.57 (4.3)	17.6 (36.4)

Shares (mean over clean processes)

curve	scan / collection	collection / cold	build / cold	selection / cold
`icv1-f2m53-tm56619371-dac20a85`	0.974	0.560	0.303	0.058
`icv1-f2m59-tm943548413-98844ecc`	0.975	0.555	0.292	0.064
`icv1-f2m61-t158598901-ab42b6c5`	0.989	0.815	0.131	0.023

Limits

- The probes' overhead is not measured: no same-binary control ran.
- The v3 figures come from a different binary, day and stage boundary.
- Seven or eight processes a size give means, not intervals.